THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A basic oxide reacts with an acid to give: 1

(a) salt and hydrogen **(b)** only salt
(c) salt and water **(d)** only water

2. The pH of a mixture of 10 mL 0.1 M HCl and 10 mL 0.1 M NaOH is about: 1

(a) 1 **(b)** 13 **(c)** 7 **(d)** 4

3. The number of water molecules in $CaSO_4 \cdot 2\,H_2O$ changing to plaster of Paris lost is: 1

(a) 2 **(b)** 1.5 **(c)** 0.5 **(d)** 1

4. The brine used in chlor-alkali process is: 1

(a) aqueous sodium hydroxide **(b)** aqueous sodium chloride
(c) molten salt **(d)** dilute HCl

5. The pH of soil is 4.5. To improve it, the farmer should add: 1

(a) quick lime **(b)** ammonium chloride
(c) vinegar **(d)** dilute HCl

6. The reaction $Na_2CO_3 + 10\,H_2O \longrightarrow Na_2CO_3 \cdot 10\,H_2O$ is: 1

(a) crystallisation **(b)** neutralisation **(c)** displacement **(d)** decomposition

7. Soda-acid fire extinguisher releases: 1

(a) hydrogen **(b)** oxygen **(c)** nitrogen **(d)** carbon dioxide

8. Two solutions A and B have pH 2 and 6. A is more acidic than B by a factor of: 1

(a) 10000 **(b)** 400 **(c)** 100 **(d)** 4

9. The anhydrous copper sulphate is used to detect: 1

(a) water **(b)** oxygen **(c)** hydrogen **(d)** carbon dioxide

10. In $CuSO_4 \cdot 5\,H_2O$ on heating the blue colour is lost because: 1

(a) copper is oxidised **(b)** water of crystallisation is lost
(c) sulphate decomposes **(d)** copper is reduced

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How does Mg(OH)2 in an antacid act in an acidic stomach? Write the equation. 2

12. Why does the colour of anhydrous copper sulphate change on adding water? Give the reason. 2

13. Write the balanced equation for the reaction of plaster of Paris with water. Name the product. 2

14. Why do we get hydrogen with HCl and zinc but not with HNO_3 and zinc in the usual lab test? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why aqueous solutions of acids and bases conduct electricity. 3

16. Compare the properties of washing soda and baking soda: formula, effect of heat and use. 3

17. Why do pickles and curd turn dull on keeping in brass vessels? 3

OR

Why does the colour of blue crystals of copper sulphate change on heating and why does it return on adding water? 3

18. Calculate the number of water molecules lost when gypsum becomes plaster of Paris and write the equation. 3

OR

Write the equations for the formation of
(a) sodium carbonate from baking soda and
(b) washing soda from sodium carbonate. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the preparation, properties and uses of baking soda, including the effect of heat. 5

OR

Explain why all salts are not neutral, with examples of acidic, basic and neutral salts. 5

20. Explain how washing soda is made. Give its properties and uses. 5

OR

Explain the uses of bleaching powder with the equation for its formation and its action on water. 5

END OF QUESTION PAPER — Total: 40 Marks

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